Opening Hour Decision Framework

Pine Script v5 — Session Classification Indicator

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Overview

This Pine Script implements the Opening Hour Decision Framework — a structured approach to classifying session character in the first 60 minutes. It enforces the no-trade window, identifies opening scenarios, and provides a session classification at 10:00 to determine whether to engage, stand down, or trade selectively for the remainder of the session.

Three Indicator Panels

Panel	Function	Output
Panel 1	Session Classification — classifies each session as Bounce Day, Trend Day, or Choppy Day at 10:00 based on zone interaction and RSI.	+1 = Bounce Day (engage). 0 = Choppy Day (selective). -1 = Trend Day (stand down).
Panel 2	Opening Hour VWAP Slope — rate of VWAP change through the opening hour. Identifies session direction and adjusts exit target expectations.	Red = steep decline. Orange = moderate. Grey = flat. Green = rising. Red background = opening spike warning.
Panel 3	Pre-Market RSI + No-Trade Window — RSI at session open with overbought warning. Yellow background during the first 5-minute no-trade window.	Red RSI line = overbought warning. Blue = oversold. Yellow background = no-trade window active.

Four Automated Alerts

1. Pre-Market RSI Warning — fires at 09:30 if RSI is above 70. Opening spike risk — observe only.

2. No-Trade Window End — fires at 09:35 (configurable). Now watching for valid setups.

3. Opening Flush — fires when price enters the zone in the first 30 minutes with RSI oversold. Highest conviction setup of the session.

4. Session Classification — fires at 10:00 with the session type label (Bounce / Trend / Choppy) and current RSI and VWAP slope values.

Session Classification Logic

Classification	Criteria	Action	Panel Value
Bounce Day	Zone touched in first 30 mins AND RSI below 40	Engage — apply three-criteria entry framework	+1 (Green)
Trend Day	Price above VWAP AND RSI above 50 AND zone not touched	Stand down — do not force entries	-1 (Red)
Choppy Day	Mixed signals — neither bounce nor	Selective — trade only highest conviction	0 (Orange)

	trend criteria clearly met	setups	

Installation

1. Open TradingView and navigate to your chart
2. Click Pine Editor at the bottom of the screen
3. Paste the full script below and click Save
4. Click Add to Chart
5. Set alerts for each of the four alert conditions

Pine Script Source Code

Copy the complete code below and paste into TradingView Pine Editor:

```
// ============================================================================
// Opening Hour Decision Framework — Pine Script v5
// Developed for: Dermot Greally | dermotgreally.github.io
// Platform: TradingView Pine Script v5
// Timeframe: 1-minute charts
// ============================================================================
// OVERVIEW:
// This script implements the Opening Hour Decision Framework — a structured
// approach to classifying session character in the first 60 minutes of trading.
// It identifies three opening scenarios, enforces a no-trade window in the
// first 5 minutes, and classifies each session as Bounce, Trend, or Choppy
// by 10:00 based on VWAP zone interaction and RSI behaviour.
//
// THREE PANELS:
// Panel 1: Session Classification (Bounce / Trend / Choppy)
// Panel 2: Opening Hour VWAP Slope
// Panel 3: Pre-Market RSI Warning + No-Trade Window
//
// FOUR ALERTS:
// 1. Pre-Market RSI Warning — RSI above 70 at session open
// 2. No-Trade Window — fires at 09:30, clears at 09:35
// 3. Opening Flush — price enters zone in first 30 minutes (highest conviction)
// 4. Session Classification — fires at 10:00 with session type label
// ============================================================================

//@version=5
indicator(title="Opening Hour Decision Framework",
          shorttitle="OH Framework",
          overlay=false,
          max_labels_count=500)

// ============================================================================
// INPUTS
// ============================================================================

// Session Settings
session_open    = input.string("0930-1600", title="Regular Session Hours", group="Session")
no_trade_mins   = input.int(5, title="No-Trade Window (minutes from open)", minval=1,
maxval=15, group="Session")
classify_time   = input.string("1000", title="Session Classification Time (HHMM)",
group="Session")

// VWAP Settings
slope_lookback  = input.int(10, title="VWAP Slope Lookback (minutes)", group="VWAP")
slope_steep     = input.float(-0.25, title="Steep Decline Threshold", group="VWAP")
slope_moderate  = input.float(-0.10, title="Moderate Decline Threshold", group="VWAP")

// RSI Settings
rsi_length      = input.int(14, title="RSI Length", group="RSI")
rsi_overbought  = input.float(70.0, title="Pre-Market Overbought Warning Level", group="RSI")
rsi_oversold    = input.float(40.0, title="RSI Oversold Threshold (Zone Entry)", group="RSI")

// Display Settings
show_classify   = input.bool(true, title="Show Session Classification Panel", group="Display")
```

```
show_slope      = input.bool(true, title="Show Opening Hour VWAP Slope Panel",
group="Display")
show_warning    = input.bool(true, title="Show Pre-Market RSI Warning Panel", group="Display")
show_alerts     = input.bool(true, title="Enable Alerts", group="Display")

// ==============================================================================
// CALCULATIONS
// ==============================================================================

// --- VWAP ---
vwap_val  = ta.vwap(hlc3)

// --- VWAP Standard Deviation Bands ---
vwap_sd   = ta.stdev(close, slope_lookback)
lower_1sd = vwap_val - vwap_sd
lower_2sd = vwap_val - (2 * vwap_sd)
upper_1sd = vwap_val + vwap_sd
upper_2sd = vwap_val + (2 * vwap_sd)

// --- VWAP Slope ---
vwap_slope = vwap_val - vwap_val[slope_lookback]

// --- RSI ---
rsi_val = ta.rsi(close, rsi_length)

// --- Session Timing ---
// Detect regular session start (09:30)
is_session        = not na(time(timeframe.period, session_open))
session_start     = is_session and not is_session[1]

// Minutes since session open
var int mins_since_open = 0
if session_start
    mins_since_open := 0
else if is_session
    mins_since_open := mins_since_open + 1

// No-trade window (first 5 minutes)
in_no_trade_window = is_session and mins_since_open < no_trade_mins

// Classification window (by 10:00 = 30 minutes in)
in_classify_window = is_session and mins_since_open <= 30

// --- Zone Detection ---
in_bounce_zone    = close < lower_1sd and close > lower_2sd
below_2sd         = close < lower_2sd
above_vwap        = close > vwap_val

// --- Opening Scenario Detection ---
// Track whether price has touched zone in first 30 minutes
var bool zone_touched_opening = false
var bool trend_day_developing = false
var string session_type = "Unclassified"

if session_start
    zone_touched_opening := false
    trend_day_developing := false
    session_type := "Unclassified"

if in_classify_window and (in_bounce_zone or below_2sd)
    zone_touched_opening := true

if in_classify_window and above_vwap and rsi_val > 50
    trend_day_developing := true

// Classify session by 10:00 (30 minutes in)
is_classify_moment = is_session and mins_since_open == 30

if is_classify_moment
    if zone_touched_opening and rsi_val < rsi_oversold
        session_type := "BOUNCE DAY"
    else if trend_day_developing and not zone_touched_opening
        session_type := "TREND DAY"
    else
        session_type := "CHOPPY DAY"

// Opening Spike Detection
```

```
// Price near or above +2SD in first 10 minutes with overbought RSI
opening_spike = is_session and mins_since_open < 10 and close > upper_1sd and rsi_val >
rsi_overbought

// Opening Flush Detection
// Price enters zone in first 30 minutes on oversold RSI — highest conviction
opening_flush = in_classify_window and in_bounce_zone and rsi_val < rsi_oversold and not
in_no_trade_window

// ============================================================================
// PANEL 1 — SESSION CLASSIFICATION
// ============================================================================

// Plot session type as a normalised value:
// +1 = Bounce Day (engage — zone touched, RSI oversold)
// 0  = Choppy Day (selective — mixed signals)
// -1 = Trend Day (stand down — price above VWAP, RSI elevated)

classify_value = session_type == "BOUNCE DAY" ? 1.0 :
                 session_type == "TREND DAY"  ? -1.0 : 0.0

classify_color = session_type == "BOUNCE DAY" ? color.new(color.green, 20) :
                 session_type == "TREND DAY"  ? color.new(color.red, 20)   :
                 color.new(color.orange, 40)

plot(show_classify ? classify_value : na,
     title="Session Classification",
     style=plot.style_histogram,
     color=classify_color,
     linewidth=3)

hline(1,  "Bounce Day",  color=color.new(color.green, 60), linestyle=hline.style_dashed)
hline(0,  "Choppy Day",  color=color.new(color.orange, 60), linestyle=hline.style_dashed)
hline(-1, "Trend Day",   color=color.new(color.red, 60), linestyle=hline.style_dashed)

// Add label at classification moment
if is_classify_moment and show_classify
    label_color = session_type == "BOUNCE DAY" ? color.new(color.green, 20) :
                  session_type == "TREND DAY"  ? color.new(color.red, 20)   :
                  color.new(color.orange, 20)
    label.new(bar_index, classify_value + 0.2,
              "10:00 CLASSIFICATION\n" + session_type + "\nRSI: " +
str.tostring(math.round(rsi_val, 1)),
              color=label_color,
              textcolor=color.white,
              style=label.style_label_down,
              size=size.small)

// Opening flush label
if opening_flush and not opening_flush[1] and show_classify
    label.new(bar_index, 1.2,
              "OPENING FLUSH\nHighest conviction setup\nRSI: " +
str.tostring(math.round(rsi_val, 1)),
              color=color.new(color.blue, 20),
              textcolor=color.white,
              style=label.style_label_down,
              size=size.small)

// ============================================================================
// PANEL 2 — OPENING HOUR VWAP SLOPE
// ============================================================================

slope_color = vwap_slope < slope_steep    ? color.new(color.red, 20)    :
              vwap_slope < slope_moderate  ? color.new(color.orange, 20) :
              vwap_slope < 0               ? color.new(color.gray, 40)   :
              color.new(color.green, 20)

plot(show_slope ? vwap_slope : na,
     title="Opening Hour VWAP Slope",
     style=plot.style_histogram,
     color=slope_color,
     linewidth=2)

hline(slope_steep,    "Steep Threshold",    color=color.new(color.red, 50),
linestyle=hline.style_dashed)
hline(slope_moderate, "Moderate Threshold", color=color.new(color.orange, 50),
linestyle=hline.style_dashed)
```

```
hline(0,              "Zero Line",          color=color.new(color.white, 50),
linestyle=hline.style_solid)

// Opening spike warning on slope panel
bgcolor(opening_spike ? color.new(color.red, 85) : na, title="Opening Spike Warning")

// ============================================================================
// PANEL 3 — PRE-MARKET RSI WARNING + NO-TRADE WINDOW
// ============================================================================

// Plot RSI in this panel — colour coded for warning conditions
rsi_color = rsi_val > rsi_overbought     ? color.new(color.red, 20)    :
            rsi_val < rsi_oversold        ? color.new(color.blue, 20)   :
            color.new(color.gray, 40)

plot(show_warning ? rsi_val : na,
     title="RSI",
     style=plot.style_line,
     color=rsi_color,
     linewidth=2)

hline(rsi_overbought, "Overbought Warning", color=color.new(color.red, 50),
linestyle=hline.style_dashed)
hline(50,             "RSI Midline",        color=color.new(color.white, 50),
linestyle=hline.style_solid)
hline(rsi_oversold,   "Oversold Entry",     color=color.new(color.blue, 50),
linestyle=hline.style_dashed)

// No-trade window background
bgcolor(in_no_trade_window ? color.new(color.yellow, 85) : na, title="No-Trade Window")

// Pre-market overbought warning label at session open
if session_start and rsi_val > rsi_overbought and show_warning
    label.new(bar_index, rsi_val + 3,
              "PRE-MARKET WARNING\nRSI: " + str.tostring(math.round(rsi_val, 1)) + "\nOpening
spike risk — observe only",
              color=color.new(color.red, 20),
              textcolor=color.white,
              style=label.style_label_down,
              size=size.small)

// No-trade window end label
if mins_since_open == no_trade_mins and is_session and show_warning
    label.new(bar_index, 50,
              "NO-TRADE WINDOW ENDS\nNow watching for valid setups",
              color=color.new(color.blue, 20),
              textcolor=color.white,
              style=label.style_label_up,
              size=size.small)

// ============================================================================
// ALERTS
// ============================================================================

if show_alerts

    // Alert 1: Pre-market RSI warning at session open
    alertcondition(session_start and rsi_val > rsi_overbought,
                   title="Pre-Market RSI Warning",
                   message="OPENING HOUR WARNING — RSI overbought at session open (" +
str.tostring(math.round(rsi_val, 1)) + "). Opening spike risk. Do not chase green candles.
Observe only.")

    // Alert 2: No-trade window end
    alertcondition(mins_since_open == no_trade_mins and is_session,
                   title="No-Trade Window Ends",
                   message="NO-TRADE WINDOW ENDS — First 5 minutes complete. Now watching for
valid zone, RSI and BEP setups.")

    // Alert 3: Opening flush detected
    alertcondition(opening_flush and not opening_flush[1],
                   title="Opening Flush — High Conviction Setup",
                   message="OPENING FLUSH DETECTED — Price has entered the -1SD to -2SD zone
in the opening 30 minutes with RSI oversold. Highest conviction setup of the session. Watch
for BEP confirmation.")

    // Alert 4: Session classification at 10:00
```

```
    alertcondition(is_classify_moment,
                   title="Session Classification — 10:00",
                   message="SESSION CLASSIFIED AT 10:00 — " + session_type + " | RSI: " +
str.tostring(math.round(rsi_val, 1)) + " | VWAP Slope: " + str.tostring(math.round(vwap_slope,
2)))

// ============================================================================
// NOTES FOR CALIBRATION
// ============================================================================
// After adding to your chart, calibrate:
//
// 1. session_open — adjust to your exchange session hours if not NYSE/NASDAQ
//
// 2. no_trade_mins — default 5 minutes. Increase to 10 if you prefer more
//    price discovery before watching for setups.
//
// 3. slope_steep / slope_moderate — calibrate against your validated dataset.
//    The default -0.25 / -0.10 thresholds are starting points.
//
// 4. rsi_overbought / rsi_oversold — defaults are 70 and 40, consistent with
//    the validated framework. Do not change without data support.
//
// 5. Session Classification logic uses zone touch + RSI to classify by 10:00.
//    On your instrument, verify that "BOUNCE DAY" classifications correlate
//    with actual bounce setups before relying on the label.
// ============================================================================
```